

Ashvin Anilkumar

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Summary

Graduate student specializing in Robotics and AI, passionate about advancing embodied intelligence through robot learning and perception. Experienced in developing autonomous systems that integrate reinforcement learning with real-world robotic control. Research interests include scalable perception, learning-based control, and sim-to-real deployment for navigation and manipulation in complex human environments.

Education

University of Wisconsin-Madison, Master of Science in Electrical and Computer Engineering December 2026

- *Relevant Coursework*: Computer Vision, Topics in Advanced Robotics, High-performance Computing, Optimization.

Government Engineering College, Barton Hill, Trivandrum, Bachelor of Technology in Electrical and Electronics Engineering July 2023

- GPA: 8.9/10 (First class with distinction)
- *Additional Coursework*: Minor in Information Technology

Research & Academic Projects

Multiagent Coordination and Decision Making (RoboCup) December 2025 – Present

- Conducting a multi-agent, real-time learning framework using reinforcement learning and robot learning for the International RoboCup Tournament under the guidance of Prof. Josiah P. Hanna.
- Supported in changing the physics engine from using **Isaac Sim's PhysX** to **Mujoco** for training the robots.
- Expected to use **Isaac Labs** for the robot learning phase of the project and achieve a great sim-to-real transfer. The work is in progress.

University Rover Challenge: Obstacle Detection and Avoidance Sep 2025 – Dec 2025

- Implemented a depth-based obstacle detection and reactive control using stereo vision for safe traversal using Luxonis OAK-D W PoE depth camera, which supports **DepthAI**.
- Developed a real-time perception and mapping framework for autonomous navigation on unstructured terrain using **ROS2**.
- Worked on implementing a ground detection algorithm using **RANSAC**.
- Implemented a simulation of the obstacle avoidance algorithm using **Gazebo**.

Work Experience

Engineer, Backend Development(AI/ML), Qburst Technologies - Trivandrum, India Aug 2023 – Aug 2025

AI Search Using RAG & Agentic Flow

- Developed an AI search bar using **FastAPI** by integrating Retrieval Augmented Generation(RAG) and Natural language to SQL conversion using **Langchain & Llamaindex**.
- Built an optimized text embeddings in **Postgres** and hosted Few-Shot Examples in **MongoDB**.
- Optimized the system for improved response speed and optimized token usage by more than **56%**.
- Increased the accuracy of search results by **90%** and expanded the range of questions it can handle by **85%**.

AI Report Reviewer System

- Formulated an AI-powered system for automated evaluation and structured feedback on technical reports using **Langchain** and **FastAPI**.
- Designed an Agentic flow that included a custom scoring algorithm to assess document structure, clarity, and relevance.
- Achieved **89%** alignment with human ratings, improved evaluation accuracy by **35%** and reduced review costs by **67%**.

Skills & Extracurricular

Robotics: ROS2, Isaac Sim, Mujoco, Gazebo

AI/ML: PyTorch, Scikit-Learn, LangChain

Tools: FastAPI, Docker, Kubernetes, Git, Linux

Hobbies: Piano, Weightlifting, Badminton, Photography

Research: Robot learning, Perception

Programming: Python, C++ , Rust, JavaScript, Cuda

Hardware: Raspberry Pi, STM32, Arduino, ESP32

Databases: MongoDB, Postgres, MariaDB, MySQL